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Anomaly Detection on Heterogeneous Graphs for the Italian Public Administration

Laboratorio di Statistica con le Aziende 2025

Project developed in collaboration with **RE-ACT**

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1 Introduction

This work is part of the *Laboratorio di Statistica con le Aziende 2025* competition, promoted by the Department of Statistical Sciences of the Università di Padova, with the aim of fostering the integration between academic training and real-world applications in business and institutional contexts.

Within this framework, the proposed solution aims to address a problem introduced by the RE-ACT research center, an organization specialized in the use of innovative data analysis methods for the prevention of criminal phenomena and anomalous behaviors in both public and private contexts. In particular, the problem addressed concerns the identification of anomalous behaviors within the Italian Public Administration, with specific reference to public procurement and contract award processes. The project is based on the development of an unsupervised *anomaly detection* model capable of identifying anomalous *patterns* within Public Administration relationships, represented through a heterogeneous graph. The proposed approach combines the construction of a multi-relational graph with the application of *Graph Neural Network* techniques for the automatic identification of potentially anomalous behaviors.

The use of unsupervised models is particularly appropriate and necessary in this scenario, since anomalous cases are not known in advance and the primary objective is to identify unusual or unexpected relational configurations within a complex system. The developed model therefore aims to combine both structural and attributive information, leveraging both the characteristics of individual entities and the relationships between them, in order to construct an interpretable anomaly score that can be useful in investigative and prevention contexts.

2 Heterogeneous Graph of the Italian Public Administration

The analysis proposed in this work is based on the construction of a heterogeneous graph representing the relationships among the main actors involved in the procurement and contract award processes of the Italian public administration. The use of a graph-based representation makes it possible to naturally model the complexity of the Public Administration system, characterized by numerous interconnected entities and relationships of different nature.

Unlike traditional graphs, where all nodes and relationships are of the same type, a heterogeneous graph allows the representation of multiple types of entities and multiple types of relationships among them. This approach is particularly suitable for the analyzed context, in which different institutional and operational actors coexist, each characterized by specific roles and attributes [8].

Formally, the heterogeneous graph can be defined as:

$$G = (V, E, \phi, \psi)$$

where:

- V represents the set of nodes

- E represents the set of edges
- $\phi : V \rightarrow \mathcal{T}_V$ is the function associating each node with its corresponding type
- $\psi : E \rightarrow \mathcal{T}_E$ is the function associating each edge with its corresponding type

Within the context of the Italian Public Administration, the constructed graph includes four main node types:

- SA (Contracting Authorities): public entities that publish procurement notices
- CIG (Tender Identification Codes): public tenders or procurement procedures
- AGZ (Awards): outcomes of public procurement procedures
- AG (Awardees): economic operators winning the tenders

These entities are connected through different types of relationships representing the real interactions occurring within the administrative process:

- $SA \rightarrow CIG$ (**publishes**): a contracting authority publishes a tender
- $CIG \rightarrow AGZ$ (**has_award**): a tender produces one or more awards
- $AGZ \rightarrow AG$ (**awarded_to**): an award is assigned to one or more economic operators
- $CIG \rightarrow CIG$ (**linked_to**): relationships between connected tenders

In order to enable more effective information propagation within the model, additional explicit inverse relationships were introduced exclusively for algorithmic purposes:

- $CIG \rightarrow SA$ (**rev_publishes**)
- $AGZ \rightarrow CIG$ (**rev_has_award**)
- $AG \rightarrow AGZ$ (**rev_awarded_to**)

This structure allows the model to learn information both in the forward and reverse directions, improving its ability to capture complex relational patterns.

Each node is also associated with a set of numerical attributes describing its observable characteristics. The nature and dimensionality of these attributes depend on the specific node type considered, reflecting the different information available for each entity within the system.

The resulting graph exhibits particularly large dimensions, making scalable graph learning techniques necessary. Additional details regarding the number of nodes and edges are shown in Figure 1.

From an interpretative perspective, the graph can be viewed as a network representing the functioning of the Public Administration: contracting authorities publish tenders, which are awarded to economic operators through procurement awards. Through this structure, it becomes possible to observe recurring patterns, frequent relationships, and atypical behaviors.

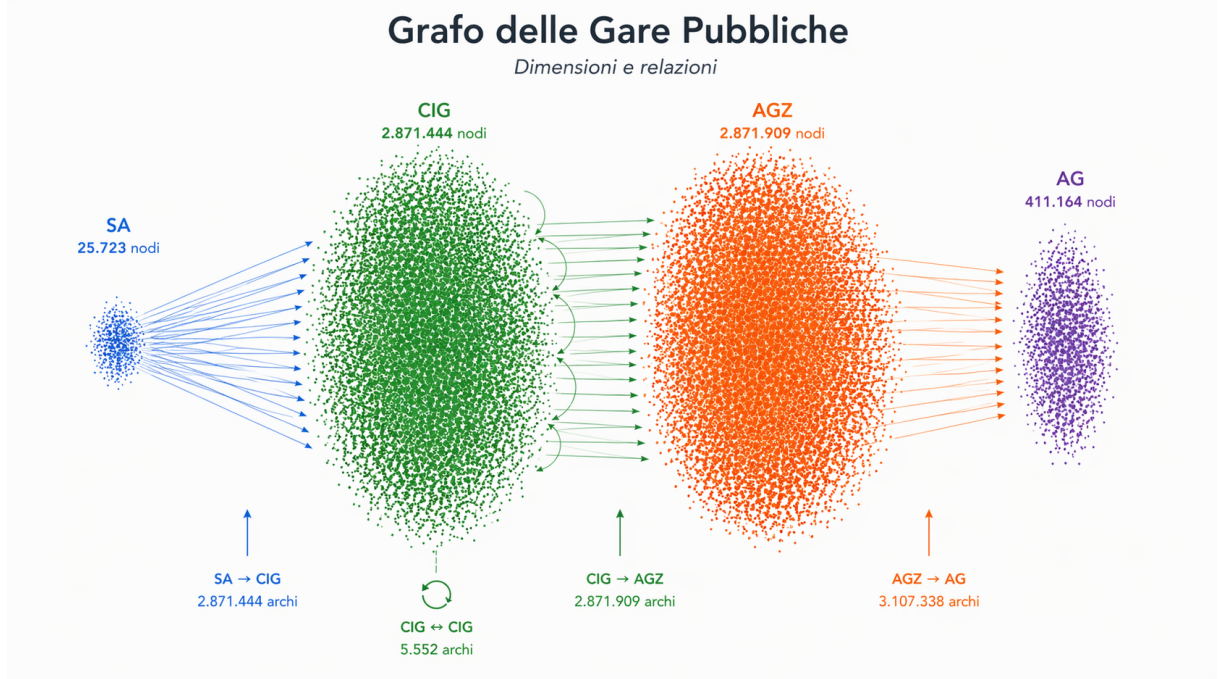


Figure 1: Graph representation

The use of a heterogeneous graph therefore enables the simultaneous integration of structural and attributive information, providing a comprehensive representation of the analyzed system. This representation constitutes the foundation for the application of Graph Neural Network-based anomaly detection techniques, described in the following sections.

2.1 Data Cleaning

The *preprocessing* phase aimed to transform heterogeneous and noisy administrative data into coherent numerical matrices suitable for tensor-based training of the heterogeneous graph *anomaly detection* model. In particular, the *preprocessing* procedure involved four main *datasets*: contracting authorities (SA), public tenders (CIG), awards (AGZ), and awardees (AG).

The first stage involved data cleaning, duplicate removal, handling of *missing* values, and standardization of numerical variables. Categorical variables were converted using *one-hot encoding* (indicator variables), while boolean variables were transformed into binary numerical format. Furthermore, for numerical variables containing missing values, explicit indicators were introduced to signal their absence, allowing the model to distinguish between observed and imputed values.

Subsequently, a relational filtering process was applied to the four tables, retaining only the entities for which it was possible to reconstruct the entire chain:

$$SA \rightarrow CIG \rightarrow AGZ \rightarrow AG$$

This filtering procedure was performed through iterative key intersection, removing all incomplete or inconsistent observations and ensuring the structural consistency of the final graph.

Finally, a progressive numerical identifier was assigned to each node type, while the original textual information was stored in external dictionaries. In this way, the final matrices contain exclusively numerical variables, ready to be converted into tensors, while the identifying information is preserved separately to guarantee interpretability of the results.

The final output of the preprocessing phase consists of a set of coherent and relationally compatible numerical datasets, ready for the construction of the heterogeneous graph and the training of the anomaly detection model.

A detailed description of the data cleaning procedures and covariate construction process is available in the dedicated document [18].

3 Anomaly Detection on Heterogeneous Graphs

Anomaly detection represents a fundamental field within statistics and machine learning, aimed at identifying observations that significantly deviate from expected behavior. In complex application contexts such as the Public Administration, the identification of anomalies can provide valuable insights for monitoring, prevention, and risk analysis activities.

In traditional settings, anomaly detection is typically applied to tabular data, where each observation is described by a set of variables. However, this approach becomes limited when the relevant information concerns not only the individual characteristics of entities, but also the relationships among them. In complex systems, anomalous behaviors may emerge not from the characteristics of a single subject, but rather from the way it interacts with other entities.

For this reason, recent years have seen the growing adoption of graph-based anomaly detection techniques [6, 7, 9], which allow the simultaneous analysis of both node attributes and relational structures. This approach is particularly suitable for contexts such as the Public Administration, where relationships among institutions, tenders, and economic operators represent a central source of information.

In the case of heterogeneous graphs, anomaly identification becomes even more complex, since the involved entities are of different nature and the relationships may carry different meanings [1, 2]. Within this framework, it is possible to distinguish three main categories of anomalies:

- attributive anomalies, involving unusual values in node characteristics
- structural anomalies, emerging from atypical connection patterns
- contextual anomalies, occurring when a behavior is anomalous with respect to its relational context

Within the context of the Italian Public Administration, these three categories assume particularly relevant meanings. An attributive anomaly may correspond, for example, to a tender characterized by unusual economic or procedural features. A structural anomaly may emerge when certain economic operators appear to be connected in an unusual way to specific contracting authorities. A contextual anomaly may arise when an apparently normal behavior becomes anomalous within the relational context in which it occurs.

In this work, the objective is not to automatically identify illegal or irregular behaviors, but rather to detect atypical configurations deserving further investigation. From this perspective, anomaly is interpreted as a statistical deviation from ordinary patterns, highlighting situations that require qualitative analysis.

This distinction is particularly important in the public sector, where interpretability of results represents a fundamental requirement. The model developed in this project is designed to provide an interpretable anomaly score, allowing analysts to understand which elements contribute to identifying atypical behavior.

A central aspect of the proposed approach is the interpretability of the results. In particular, the model makes it possible to distinguish between different components of the anomaly score, separating the contribution of node attributes from that of the graph structure. This distinction allows for understanding whether an anomaly originates from unusual attributes, anomalous relationships, or a combination of both.

4 Model

This section describes the proposed model for anomaly detection on heterogeneous graphs. The model follows a heterogeneous graph autoencoder architecture, composed of a relation-aware encoder and two distinct decoders: an attributive decoder and a relation-specific structural decoder [3, 4].

The objective of the model is to learn a latent representation for each node that summarizes both its attributive characteristics and its relational structure with neighboring nodes. Once these synthetic representations (*embeddings*) have been constructed, the model attempts to reconstruct both the original attributive and structural information associated with each node [5]. Reconstruction difficulties are then exploited to identify anomalous configurations. The larger the discrepancy between the original information and structure of the nodes and their reconstruction, the higher the assigned anomaly score.

4.1 Relation-aware encoder

The encoder learns latent node *embeddings* through two layers of heterogeneous *message passing*. For each node type $t \in \mathcal{T}_V$ (SA, CIG, AGZ, AG), we define the input *feature* matrix as:

$$X^{(t)} \in \mathbb{R}^{N_t \times d_t}$$

where N_t is the number of nodes of type t and d_t is the dimensionality of the original features.

The proposed encoder consists of two consecutive layers:

$$H^{(1)} = \text{ReLU}(\text{Conv}_1(X))$$

$$Z = \text{Conv}_2(H^{(1)})$$

For each node i , a latent vector representation $z_i \in \mathbb{R}^d$ is obtained, encoding not only the

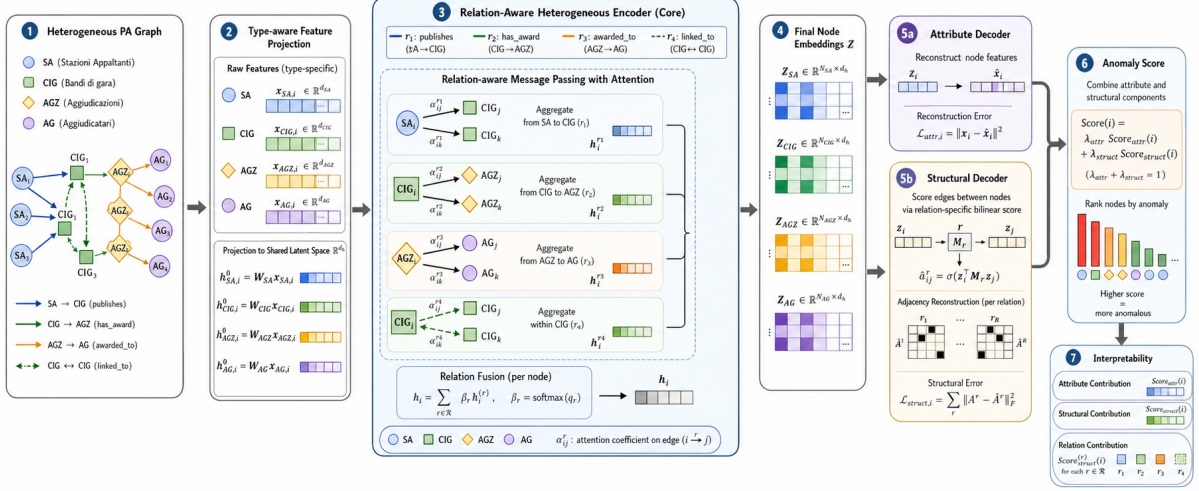


Figure 2: Proposed anomaly detection model for heterogeneous graphs

information of the node itself, but also the information of neighboring nodes and the relationships in which it is involved.

To achieve this, a convolution operator, Conv, is defined. For each node i , it computes a latent representation h_i by aggregating the representations suggested by all incoming edges associated with the target node i , separated by relation type, denoted as $h_i^{(r)}$.

$$h_i = \sum_{r \in \mathcal{R}(i)} \beta_r h_i^{(r)}$$

where:

- $\mathcal{R}(i)$ is the set of relations in which node i is involved as a target node
- the parameters β_r are defined as $\beta_r = \frac{\exp(\gamma_r)}{\sum_{r'} \exp(\gamma_{r'})}$, where the parameters γ_r are learnable and relation-specific

For example, a CIG node i is involved as a target node in the relations **publishes**, **rev_has_award**, and **linked_to**; its representation produced by this encoder will therefore be an aggregation of the representations suggested by all relations of these three types in which it is involved:

$$h_i = \beta_{\text{publishes}} h_i^{\text{publishes}} + \beta_{\text{rev_has_award}} h_i^{\text{rev_has_award}} + \beta_{\text{linked_to}} h_i^{\text{linked_to}}$$

The representations suggested by the different relations, $h_i^{(r)}$, are obtained by aggregating the information from all source nodes connected to the target node i through relation r (weighted using a relation-aware attention mechanism), together with the information contained in the target node itself:

$$h_i^{(r)} = \sum_{j \in \mathcal{N}_r(i)} \alpha_{ij}^{(r)} W_r h_j + W_{\text{self}} h_i$$

where:

- $\alpha_{ij}^{(r)}$ is an attention coefficient determining how relevant the information transmitted from source node j to target node i through relation r is
- W_r is a learnable transformation matrix specific to relation r for the source node
- h_j is the input representation of the source node
- W_{self} is a learnable transformation matrix for the target node
- h_i is the input representation of the target node being updated

To compute the attention coefficients $\alpha_{ij}^{(r)}$, the algorithm relies on attention scores $e_{ij}^{(r)}$:

$$e_{ij}^{(r)} = \text{LeakyReLU} \left(a_r^T [K_r h_i \parallel W_r h_j] \right)$$

where:

- $\parallel$ denotes concatenation
- a_r is a learnable parameter vector
- K_r is a learnable transformation matrix for the target node specific to relation r

The attention coefficients are normalized through a softmax function:

$$\alpha_{ij}^{(r)} = \frac{\exp(e_{ij}^{(r)})}{\sum_{k \in \mathcal{N}_r(i)} \exp(e_{ik}^{(r)})}$$

where:

- $\mathcal{N}_r(i)$ is the set of source nodes connected to node i through relation r

4.2 Attributive decoder

The attributive decoder reconstructs the original node features, $\hat{x}_i$, starting from the latent embeddings produced by the encoder, h_i . The structure simply consists of a set of Multi Layer Perceptrons composed of two linear layers, one for each node type.

For each node type t :

$$\hat{X}^{(t)} = f_t(Z^{(t)})$$

where f_t is a feed-forward neural network:

$$f_t(z) = W_2^{(t)} \text{ReLU}(W_1^{(t)} z)$$

where:

- $W_1^{(t)}$ and $W_2^{(t)}$ are two learnable parameter matrices

4.3 Structural decoder

The structural decoder assigns a score to each edge using a relation-specific bilinear function.

The score associated with an edge (u, v) is:

$$s_{uv}^{(r)} = z_u^T M_r z_v$$

where:

- M_r is a learnable parameter matrix specific to relation r
- z_u and z_v are respectively the embeddings of the source node and the target node involved

4.4 Loss

The loss function minimized during model training is composed of two components:

$$L = \alpha L_{\text{attr}} + \beta L_{\text{struct}}$$

where:

- L_{attr} is the loss function associated with the attributive component of the model
- L_{struct} is the loss function associated with the structural component of the model
- α is a hyperparameter controlling the weight of the attributive component
- β is a hyperparameter controlling the weight of the structural component

The attributive loss for each node type t is defined as:

$$L_{\text{attr}}^{(t)} = \frac{1}{N_t} \sum_{i=1}^{N_t} \|x_i^{(t)} - \hat{x}_i^{(t)}\|^2$$

which are then aggregated into the overall attributive loss:

$$L_{\text{attr}} = \sum_{t \in \mathcal{T}_V} \frac{N_t}{N} L_{\text{attr}}^{(t)}$$

where N is the total number of nodes in the graph.

Regarding the structural component, the following are used during *training*:

- positive edges
- negative edges generated through *negative sampling*

These allow the computation of the binary cross-entropy for each relation r :

$$L_{\text{struct}}^{(r)} = -\frac{1}{|E_r|} \sum_{(u,v) \in E_r} \left[y_{uv}^{(r)} \log \sigma(s_{uv}^{(r)}) + (1 - y_{uv}^{(r)}) \log(1 - \sigma(s_{uv}^{(r)})) \right]$$

where:

- E_r is the set of edges belonging to relation r
- $s_{uv}^{(r)}$ is the score assigned by the structural decoder to relation r between nodes u and v
- $y_{uv}^{(r)}$ is equal to 1 if an edge of type r exists between nodes u and v , and 0 otherwise
- $\sigma(\cdot)$ is the sigmoid function, which maps the scores into values between 0 and 1

The total structural loss is finally obtained as the sum of all the cross-entropies associated with the different relations r :

$$L_{\text{struct}} = \sum_r L_{\text{struct}}^{(r)}$$

4.5 Anomaly score

Since the objective of the algorithm is to identify anomalous entities within the graph, once the model has been trained it becomes necessary to assign a score to each node in order to quantify its atypicality with respect to the rest of the complex structure.

For each node i belonging to type t , the attributive score is computed as the average feature reconstruction error:

$$\text{Score}_{\text{attr}}(i) = \frac{1}{d_t} \sum_{k=1}^{d_t} \left(x_{ik}^{(t)} - \hat{x}_{ik}^{(t)} \right)^2$$

where:

- $x_i^{(t)}$ represents the vector of original node features
- $\hat{x}_i^{(t)}$ represents the vector reconstructed by the attributive decoder
- d_t is the feature dimensionality for node type t

For interpretability purposes, the attributive component of the anomaly score can be further decomposed by analyzing the contribution of individual variables. In particular, since the attributive score is defined as the mean squared reconstruction error, it is possible to examine the individual terms $(x_{ik}^{(t)} - \hat{x}_{ik}^{(t)})^2$, representing the discrepancy between observed and predicted values for each feature. This decomposition makes it possible to identify which variables contribute most to the anomaly of a node, providing an interpretable indication of the possible causes of atypical behavior.

The structural score is instead computed as the average error over the observed edges incident to node i . Specifically, for each edge (i, j) belonging to relation r , the following quantity is considered:

$$\text{Score}_{\text{struct}}(i) = \frac{1}{|N(i)|} \sum_{j \in N(i)} -\log \sigma \left(s_{ij}^{(r)} \right)$$

where:

- $N(i)$ is the set of nodes connected to node i

- $s_{ij}^{(r)}$ is the score produced by the structural decoder for relation r
- $\sigma(\cdot)$ is the sigmoid function

Since the score distributions may differ significantly across node types, the two components are normalized separately for each node type through a transformation based on the empirical cumulative distribution function (ECDF):

$$\widetilde{\text{Score}}_{\text{attr}}(i) = \text{ECDF}_t(\text{Score}_{\text{attr}}(i))$$

$$\widetilde{\text{Score}}_{\text{struct}}(i) = \text{ECDF}_t(\text{Score}_{\text{struct}}(i))$$

where $\text{ECDF}_t(\cdot)$ represents the empirical cumulative distribution function computed over nodes of type t .

The final anomaly score is then obtained as a linear combination of the two normalized components:

$$\text{Score}(i) = \lambda_{\text{attr}} \widetilde{\text{Score}}_{\text{attr}}(i) + \lambda_{\text{struct}} \widetilde{\text{Score}}_{\text{struct}}(i)$$

where λ_{attr} and λ_{struct} are hyperparameters controlling the weight of the attributive and structural components, respectively.

This allows the model to be used as a decision-support tool rather than as a fully automatic decision-making system, making it particularly suitable for the context of the Public Administration.

5 Results

This section analyzes the results obtained from applying the proposed model to the data described in Section 1. For validation purposes, the most anomalous cases identified by the model were cross-referenced with publicly available news and information sources, in order to find real-world evidence supporting the system’s suggestions.

5.1 Analysis of Awards (AGZ)

The final anomaly score indicates how far each individual case deviates from what the model has learned to consider “normal” within the public procurement system. This value ranges between 0 and 1: the closer the score is to 1, the more atypical the detected behavior is and the more worthy it becomes of further investigation. Table 1 reports the 10 awards with the highest anomaly scores.

Equivalent results were obtained for each type of node (SA, CIG, AG, AGZ), organized into separate rankings.

However, considering the structure of the model, the variables involved and the context of public procurement, we believe that the greatest attention should be devoted to the graph nodes representing awards, namely the AGZ nodes. Indeed, although any entity involved in the procurement process may potentially exhibit fraudulent behavior, the variables associated

Table 1: Top 10 Awards (AGZ) by Anomaly Score.

Pos.	ID AGZ	Aggiudicatario (AG)	Stazione Appaltante (SA)	Final	Attr.	Struct.
1	1064558	N/D	N/D	0.999999	0.999999	0.999999
2	2211528	IMPRESA INESISTENTE	COMUNE DI CASTELLEONE	0.999998	0.999998	0.999998
3	11934929	CONSORZIO DEI COMUNI TRENTINI	COMUNE DI VALLELAGHI	0.999997	0.999997	0.999995
4	1679785	N/D	N/D	0.999996	0.999996	0.999996
5	415034	N/D	N/D	0.999995	0.999994	0.999999
6	14862235	TIM S.P.A.	ISTITUTO SUPERIORE INDUSTRIE ARTISTICHE	0.999994	0.999993	0.999997
7	2594097	N/D	N/D	0.999993	0.999992	0.999998
8	1192361	N/D	N/D	0.999986	0.999984	0.999996
9	14510133	CENTRO STUDI ENTI LOCALI S.P.A.	COMUNE DI PADOVA	0.999928	0.999925	0.999939
10	15650666	DUAL ITALIA S.P.A.	AZIENDA MUNICIPALE GAS SPA	0.999927	0.999947	0.999845

with SA and AG nodes in the proposed graph are rarely sufficient to detect such behaviors (tax identification number, legal headquarters, publicly owned company status, etc.). Instead, the behavior and potential irregular conduct within a procurement procedure are much more likely to emerge through the variables encoded in the award-related nodes of the graph, such as award date, number of admitted offers, subcontracting, discount percentages, accelerated procedures, and similar characteristics. Consequently, our analyzes focus mainly on the results obtained for the AGZ nodes.

5.2 Standardization and Interpretation of the Reconstruction Error

Once an anomaly score has been assigned to each node, it becomes essential to understand the reasons why the model produced such values. Although the structural component of the anomaly score is difficult to analyze directly, except by examining the scores associated with each node connected to the node under investigation, anomaly identification also relies on the attributive reconstruction error of the model, which, unlike the structural component, can be analyzed in detail.

By examining this information, it immediately becomes evident that these deviations are symptomatic of:

1. **Behavioral Anomalies:** Real deviations from standard market or administrative practices.
2. **Data Entry Errors:** Incorrect manual entries (e.g., inconsistent dates, out-of-scale amounts) which nevertheless represent a risk for transparency in themselves.

Table 2 presents the details of the top 5 most anomalous variables for the award ranked first in the anomaly classification.

Table 2: Analysis of the attributive score for the Top 1 Award (AGZ ID 1064558).

Variable (Feature)	Observed Value	Predicted Value	Error
data_aggiudicazione_definitiva	276.8244	101.9566	30578.7578
data_comunicazione_esito	-0.2044	114.2491	13099.5859
ribasso_aggiudicazione	-0.3816	32.0452	1051.4929
importo_aggiudicazione	-0.0122	9.5114	90.6981
flag_subappalto	0.0000	8.4739	71.8070

As can be observed, the anomaly is driven almost entirely by significant reconstruction errors on temporal variables. The observed standardized value of 277 for the award date is completely out of scale, strongly suggesting either a data entry error or an impossible administrative timeline. The model isolated this case precisely because of these extreme temporal inconsistencies. Furthermore, it can be noted that the observed values for the remaining variables are relatively normal (close to 0 after standardization), while the model predicted much more extreme values, thereby contributing to the increase in the attributive error.

5.3 External Validation

To externally validate the model, the awards associated with the highest anomaly scores were verified through open sources (news articles, institutional databases, judicial records). The objective is not to formulate accusations, but rather to assess whether the detected signals correspond to already known phenomena, thereby reinforcing the usefulness of the model as a risk prioritization tool.

Most Relevant Cases among the Top 100 Awards

IMPRESA INESISTENTE – Comune di Castelleone (score: 0.999998). The awardee is reported as “*Non-Existent Company*”, suggesting either an unregistered or fictitious entity. This constitutes a violation of participation requirements (Legislative Decree 36/2023, art. 94) and may indicate possible fraud-related scenarios. The model assigns the maximum anomaly score in both the attributive and structural components.

RECORDATI RARE DISEASES ITALY S.R.L. – InnovaPuglia S.P.A. (score: 0.999373). InnovaPuglia has been involved in judicial investigations and administrative disputes concerning procurement procedures. The anomaly arises from the weak coherence between the role of the central purchasing body and a highly specialized pharmaceutical supplier.

SER.I.T. S.R.L. – Comune di Villafranca di Verona (score: 0,99969393). The company was subject to searches conducted by the Italian Financial Police under the order of the Verona Public Prosecutor’s Office for alleged abuse of office and fraud [11]. The case is linked to a long-standing administrative dispute (Regional Administrative Court and Council of State) with the Municipality regarding the construction of a waste treatment facility, characterized by repeated suspension measures.

MEDTRONIC ITALIA S.P.A. – Azienda Sanitaria Universitaria Friuli Centrale (score: 0,99936295). The company was mentioned in the “*Vasi Comunicanti*” investigation concerning an alleged corruption system in the supply of cardiological devices (stents and TAVI). Investigators hypothesized that sponsorships of training events could influence hospital purchasing decisions; however, the proceedings were later dismissed by the investigating judge due to the absence of objective evidence [12].

NETWEEK S.P.A. – Museo Nazionale del Cinema – Fondazione Maria Adriana Prolo (score: 0,99919677). The operator presents a critical financial situation: as of April 30, 2025, the group reported a net debt of 3.7 million euros, in addition to 13.5 million euros in overdue tax liabilities and 5.1 million euros in unpaid commercial debts [13]. Such financial fragility raises concerns regarding the sustainability of the assigned contracts.

SEVEN SOUNDS GROUP – Municipality of Tuscania (score: 0.999686). The Municipality of Tuscania was subject to investigations by ANAC and the Court of Auditors regarding suspected irregularities in procurement procedures between 2016 and 2021 [14]. Numerous below-threshold direct awards through SmartCIG procedures emerged, together with disputes concerning price determination and compliance with participation requirements by awardees.

LA SEMAFORICA S.R.L. – Comune di Tremezzina (score: 0,99966407). The operator acts within a territorial context affected by investigations conducted by the Milan Public Prosecutor’s Office into alleged corruption episodes related to the Tremezzina infrastructure project [15]. Investigators hypothesized a bribery system involving public officials aimed at facilitating the award of large infrastructure contracts within the same geographical area.

ENEL X ITALIA S.R.L. – Comune di Barete (score: 0,99853426). The case is connected to the complex administrative management of post-earthquake reconstruction, characterized by significant legal disputes. A controversy concerning the special administration of building consortia reached the Council of State [16], which confirmed the legitimacy of the municipal actions while highlighting an extremely fragmented and conflictual territorial governance structure.

ACCORD HEALTHCARE ITALIA S.R.L. – Policlinico Umberto I (score: 0,99852294). The awardee operates within a critical institutional context involving Policlinico Umberto I, which has been subject to multiple investigations concerning fraud, abuse of office, and false accounting in procurement and financial management [17]. The investigations revealed serious deficiencies in contract management and hospital safety procedures.

TIM S.P.A. – Istituto Superiore Industrie Artistiche (score: 0.999995). Inconsistencies emerge between the supplier profile (telecommunications) and the nature of the contracting authority (artistic education). The anomaly is predominantly structural, indicating an atypical and potentially non-competitive relationship.

ABC FARMACEUTICI S.P.A. – Regione Calabria (score: 0.998636). The award takes place within a regional context already involved in investigations (operation “*Sartoria*”, 2024) concerning irregularities in healthcare procurement contracts. Although no direct connection emerged, the broader context strengthens the relevance of the detected signal.

Aggiudicatari Persone Fisiche su Comuni di Piccole Dimensioni. A recurring pattern involves procurement awards assigned to individual persons in small municipalities. The model identifies these cases because they deviate from expected procurement profiles.

Concluding Remarks on the Validation

The external validation highlights three main aspects. First, the absence of publicly available evidence does not imply the absence of risk, since many irregularities remain undetected or inaccessible to public scrutiny. Second, several contexts identified by the model already present documented critical issues, suggesting the model’s ability to capture systemic vulnerabilities in addition to isolated anomalies. Third, the “*Impresa Inesistente*” case reveals limitations in the input data itself (errors or fictitious entities), which already constitute a risk factor and require upstream validation procedures.

Overall, the validation reinforces the value of the model as an operational support tool for supervisory authorities (e.g. ANAC), helping concentrate inspection and monitoring resources on the most critical cases.

6 Discussion, Limitations, and Future Developments

The proposed model is designed as a support tool for the analysis and monitoring of Public Administration processes, rather than as a fully automatic decision-making system. In particular, its primary objective is to provide a ranking of potentially anomalous configurations that may deserve further investigation by analysts, domain experts, or competent authorities. In this sense, the anomaly score should be interpreted as a statistical indicator of deviation from expected behavior, rather than as evidence of irregular or unlawful conduct.

The operational use of the model may prove useful in several activities, including the preventive monitoring of procurement procedures, the identification of atypical relational patterns between institutions and economic operators, and the prioritization of administrative inspections. Furthermore, the possibility of distinguishing between the attributive and structural components of the score provides more interpretable indications, facilitating the analysis of the possible causes underlying the detected anomaly.

Nevertheless, some caution is required when interpreting the results. The anomalies identified by the model may also originate from legitimate factors, such as specific territorial characteristics, particular administrative needs, or exceptional events. For these reasons, the model should be used as an exploratory and decision-support tool, integrated with domain expertise and qualitative analysis. In this way, the proposed approach may contribute to improving monitoring and prevention capabilities, while at the same time ensuring a responsible and informed use of anomaly detection techniques.

The work presented in this report therefore represents a first step toward the construction of a heterogeneous graph-based anomaly detection system applied to the Italian Public Administration. Although the project exhibits a methodological structure coherent with the addressed problem, it

also presents several limitations that should be explicitly acknowledged, both to ensure a correct interpretation of the results and to outline possible future developments.

Limitations

Computational resources. The size of the constructed graph, which includes millions of nodes and relationships, makes the training of large-scale neural models particularly computationally demanding. For this reason, hyperparameter optimization had to be necessarily limited. Similarly, the depth and complexity of the autoencoder were constrained to configurations compatible with the available memory and computational time limits.

Data quality and completeness. As is often the case with administrative *datasets*, the presence of missing values, encoding errors, duplicates, and semantic inconsistencies required an extensive *preprocessing* phase. In several cases, it was necessary to perform imputations, remove incomplete rows, or exclude entire variables considered excessively compromised. Consequently, the final graph also reflects a series of simplification choices that may have led to some loss of information.

Graph scope. The constructed graph was restricted to entities and relationships for which it was possible to reliably reconstruct the main path connecting contracting authorities, tenders, awards, and awardees. As a result, several potentially relevant components of the public procurement system have not yet been included in the current model.

Future Developments

These limitations naturally suggest the main directions for future developments. A first possible extension concerns the expansion of the graph structure through the introduction of new node and relationship types, in particular procedures related to *SmartCIG* contracts and subcontracting activities. Another promising direction involves the inclusion of temporal information, thereby transforming the current static graph into a dynamic structure. From a modeling perspective, future work could explore deeper architectures and more expressive decoders, together with a systematic hyperparameter optimization phase.

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